Tingwei Zhang

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Research Interests

Tingwei focuses on exploring security and privacy challenges in machine learning technologies, particularly in real-world scenarios and under adversarial conditions, to develop secure, ethical, and privacy-preserving AI systems.

Education

Cornell University 2023–Present

Ph.D. in Computer Science

– Advised by Vitaly Shmatikov

– M.S. in Computer Science, awarded en route to the Ph.D. program (December 2025)

University of Virginia (UVA) 2020–2023

B.A. in Computer Science with Minor in Statistics

– Graduated with *Highest Distinction* in Distinguished Majors Program in computer science.

– Worked with Prof. David Evans and Prof. Yuan Tian on security of machine learning projects at SRG at UVA.

Publications

Conference Papers

Tingwei Zhang, S. Fnu, R. Jha, C. Zhang, V. Shmatikov. "Adversarial hubness in multi-modal retrieval," *Proceedings of the 47th IEEE Symposium on Security and Privacy (IEEE S&P)*, San Francisco, CA, 2026.

Tingwei Zhang, C. Zhang, J. X. Morris, E. Bagdasarian, V. Shmatikov. "Self-interpreting adversarial images," *Proceedings of the 34th USENIX Security Symposium (USENIX Security),* Seattle, WA, 2025.

Tingwei Zhang*, R. Jha*, E. Bagdasarian, V. Shmatikov. "Adversarial illusions in multi-modal embeddings," *Proceedings of the 33rd USENIX Security Symposium (USENIX Security),* Philadelphia, PA, 2024. (**Distinguished Paper Award**)

C. Zhang, Tingwei Zhang, V. Shmatikov. "Adversarial decoding: generating readable documents for adversarial objectives," *Findings of the 19th Conference of the European Chapter of the Association for Computational Linguistics (EACL),* Rabat, Morocco, 2026.

S. Fnu, A. Suri, Tingwei Zhang, J. Hong, Y. Tian, and D. Evan. "SoK: Pitfalls in evaluating black-box attacks," *Proceedings of the 2nd IEEE Conference on Secure and Trustworthy Machine Learning (SaTML),* Toronto, Canada, 2024.

Preprints

Tingwei Zhang, J. X. Morris, V. Shmatikov. "How to steal reasoning without reasoning traces," *Preprint,* 2026.

Tingwei Zhang, H. Triedman, V. Shmatikov. "Deep-Research Agents Can Be Poisoned via User-Generated Content," *Preprint,* 2026.

See Google Scholar profile for a full list.

Internship

Student Researcher, Core, Google, San Jose, CA May–Aug. 2026

– Designed an activation-adapter method that extends frozen text-only LLM safety probes to image, audio, and interleaved multi-modal inputs.

SecEng Intern, AI Security, Amazon, Seattle, WA May–Aug. 2025

– Developed an automated framework to detect security vulnerabilities in audio-based AI systems, integrating with existing AGI security testing tools.

Teaching

Cornell Tech CS5434: Trustworthy AI, TA — Fall 2025
Cornell Tech CS5450: Networked and Distributed Systems, Head TA — Spring 2024
Cornell University CS2110: Object-Oriented Programming and Data Structures, TA — Fall 2023
UVA CS4774: Machine Learning, TA — Fall 2022
UVA CS4102: Algorithms, TA — Spring 2022

Honors & Awards

Distinguished Paper Award at USENIX Security — Aug. 2024
USENIX Security Student Grant, 2024 — Aug. 2024
Dean's List of Distinguished Students, College of Arts & Sciences, UVA — 2021 & 2022

Talks & Presentations

Adversarial Hubness in Multi-modal Retrieval
Conference Talk, IEEE S&P 2026 — May. 2026

Self-interpreting Adversarial Images
Conference Talk, USENIX Security Symposium 2025 — Aug. 2025

Adversarial Illusions in Multi-Modal Embeddings
Conference Talk, USENIX Security Symposium 2024 — Aug. 2024
Invited Talk, RSAC 2025 — Apr. 2025

Attacking and Defending Multi-Modal Representations
Invited Talk, University of Virginia CS, Security Seminar — Dec. 2024
Invited Talk, University of Massachusetts Amherst, AI Security Seminar — Oct. 2025

Services

Cornell CS Ph.D. Admissions Committee Student Reviewer — 2025–2026
Reviewer: NeurIPS 2026; ICML 2026; IEEE JSAIT
Artifact Evaluation Committee: USENIX Security 2026